Microsoft

Comet
A .NET Experiment

David Ortinau
Principal Program Manager
Twitter: davidortinau
GitHub: davidortinau
Twitch: davidortinau

James Clancey
Principal PM Architect
Twitter: @JTClancey
GitHub: Clancey
Twitch: Clancey



https://github.com/Clancey/Comet

Comet Concept

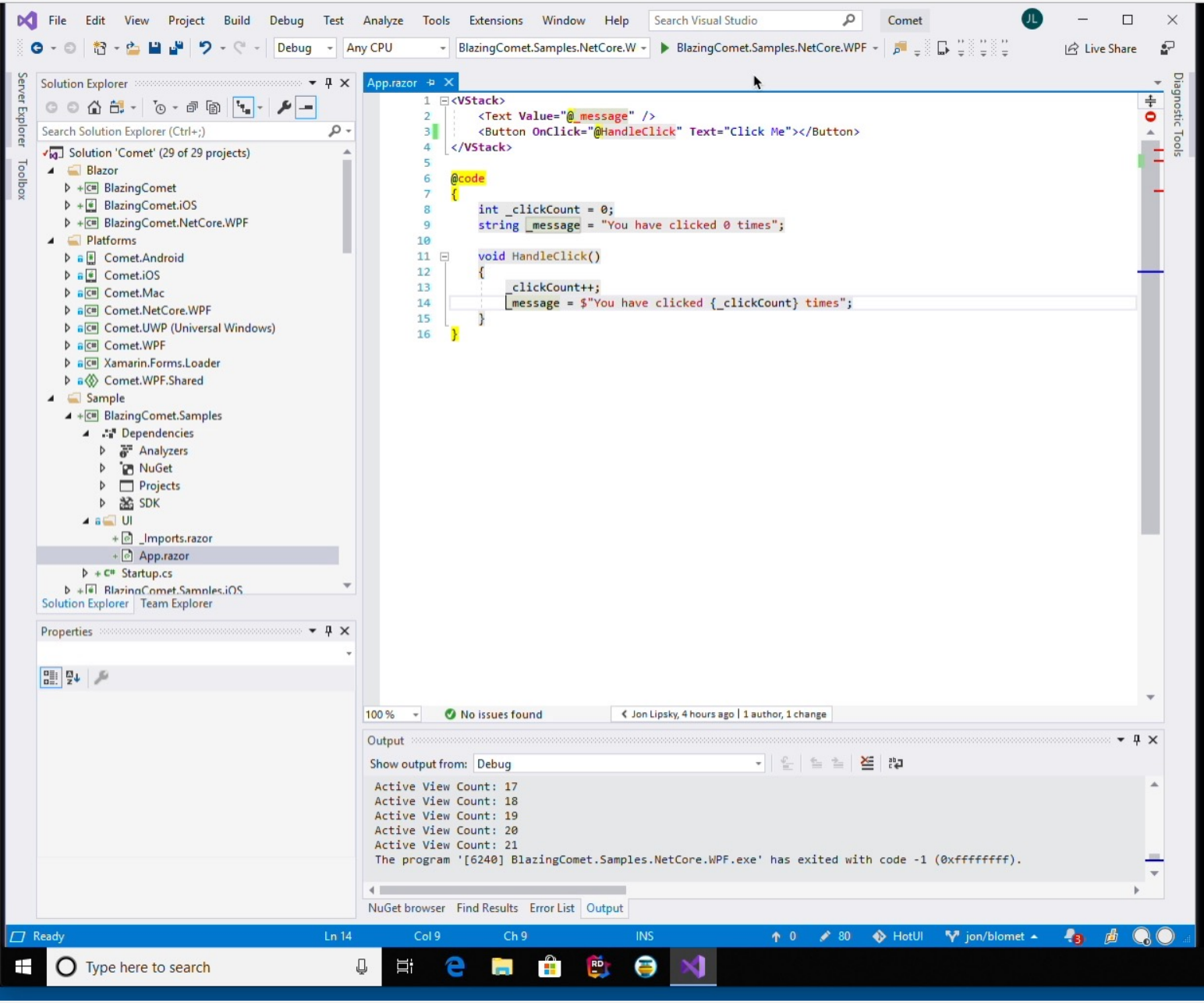
How might we fluently write UI in C#?

What would a solution look like that optimized for development productivity and was a pleasure to write?

Comet is a platform agnostic control toolkit to describe your UI and share it universally across multiple platforms.

Scenario A: Blazor + Comet

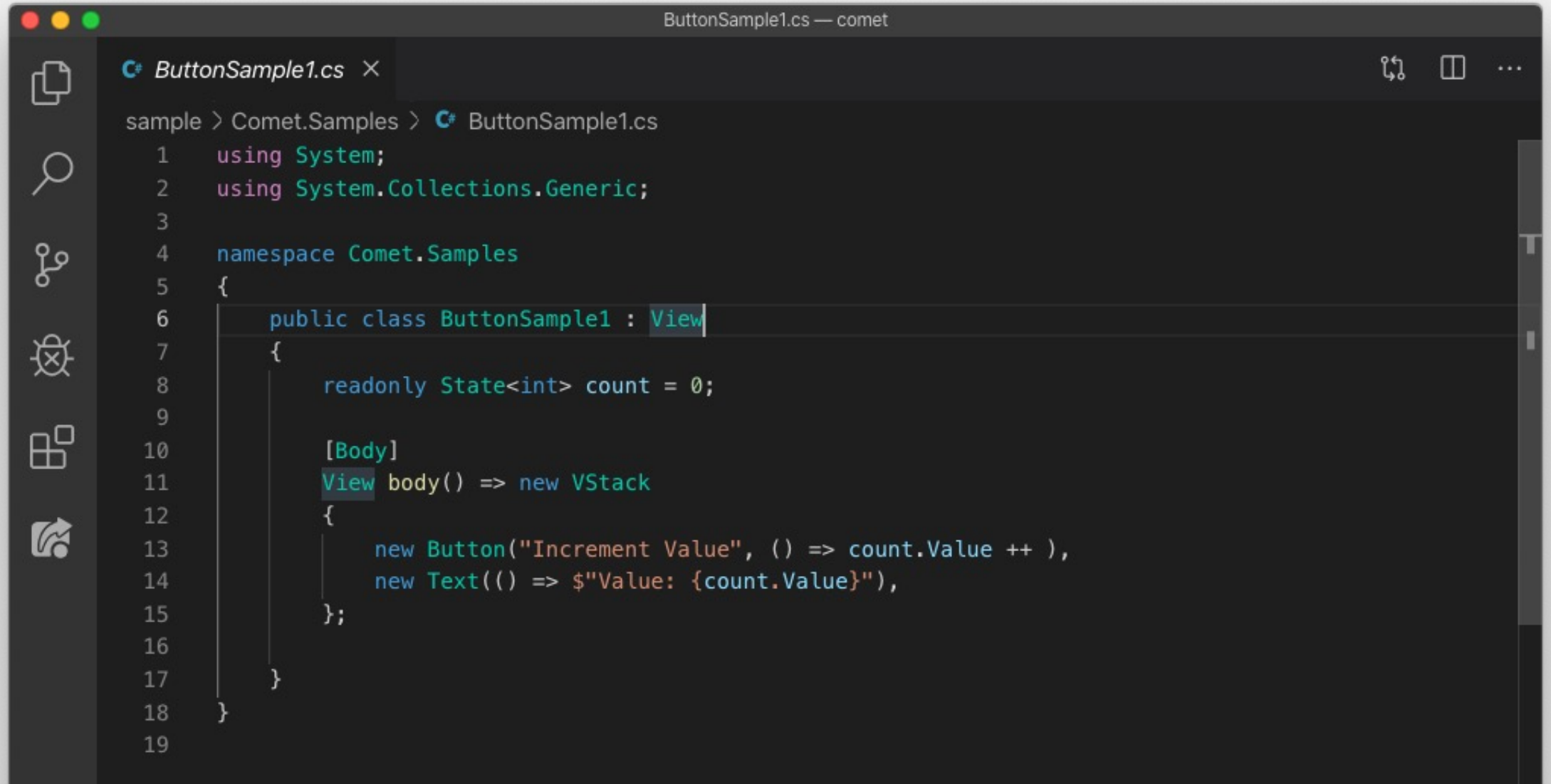
Develop with the Blazor development model and use Comet to reach native platforms outside the web browser.

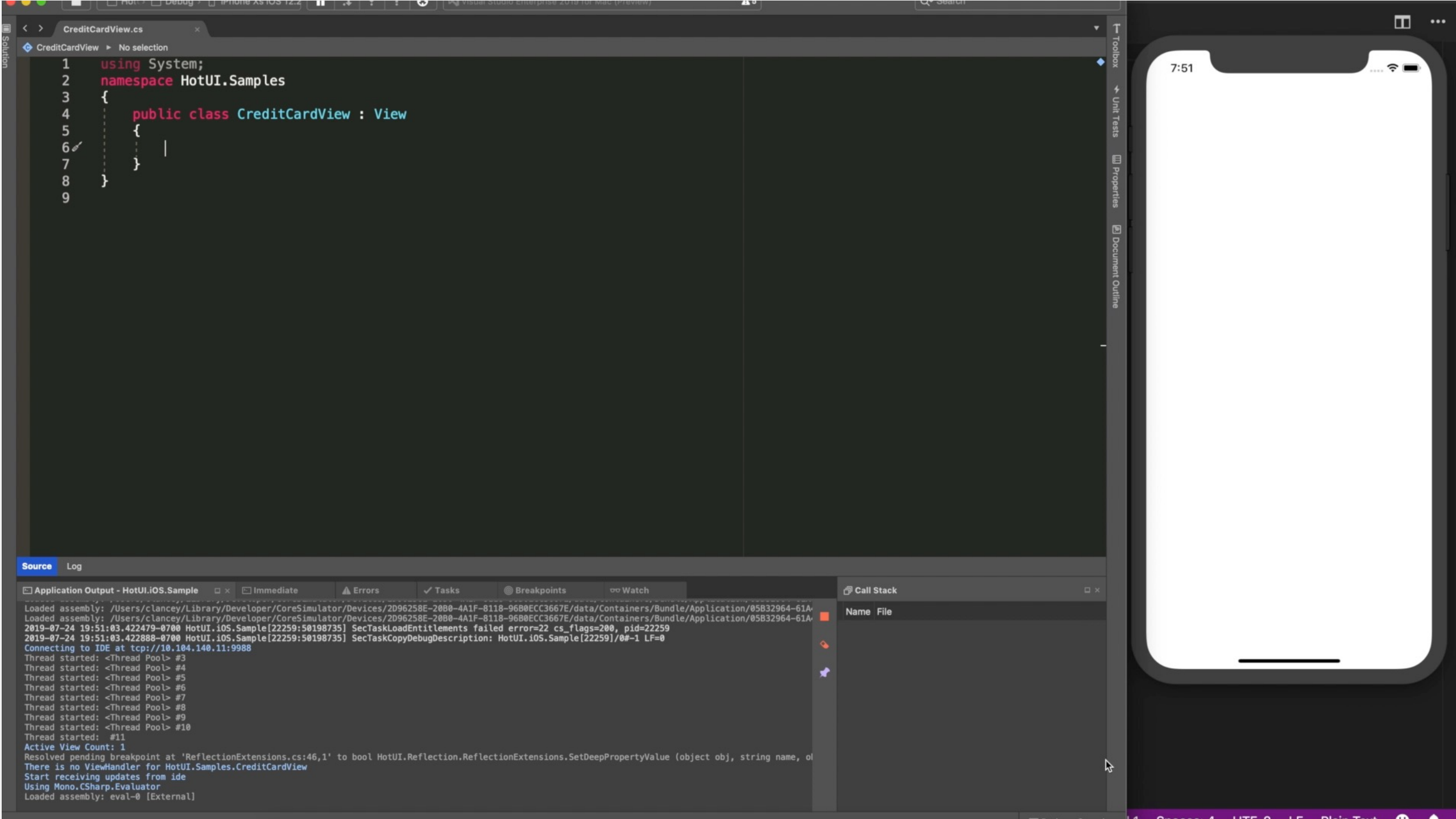

File Edit View Project Build Debug Test Analyze Tools Extensions Window Help
Search Visual Studio
Comet
Debug
Any CPU
BlazingComet.Samples.NetCore.W
BlazingComet.Samples.NetCore.WPF
Live Share
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'Comet' (29 of 29 projects)
Blazor
BlazingComet
BlazingComet.iOS
BlazingComet.NetCore.WPF
Platforms
Comet.Android
Comet.iOS
Comet.Mac
Comet.NetCore.WPF
Comet.UWP (Universal Windows)
Comet.WPF
Xamarin.Forms.Loader
Comet.WPF.Shared
Sample
BlazingComet.Samples
Dependencies
Analyzers
NuGet
Projects
SDK
UI
_Imports.razor
App.razor
Startup.cs
Solution Explorer
Team Explorer
Properties
App.razor
1 <VStack>
2 <Text Value="@_message" />
3 <Button OnClick="@HandleClick" Text="Click Me"></Button>
4 </VStack>
5
6 @code
7 {
8 int _clickCount = 0;
9 string _message = "You have clicked 0 times";
10
11 void HandleClick()
12 {
13 _clickCount++;
14 _message = $"You have clicked {_clickCount} times";
15 }
16 }
100 %
No issues found
Jon Lipsky, 4 hours ago | 1 author, 1 change
Output
Show output from: Debug
Active View Count: 17
Active View Count: 18
Active View Count: 19
Active View Count: 20
Active View Count: 21
The program '[6240] BlazingComet.Samples.NetCore.WPF.exe' has exited with code -1 (0xffffffff).
NuGet browser
Find Results
Error List
Output
Ready
Ln 14
Col 9
Ch 9
INS
HotUI
jon/blomet
Type here to search
5:35 PM
8/16/2019

Scenario B: Comet + _____

Develop with the Comet development model and use Blazor to reach the web browser, and any other platform to run natively there.

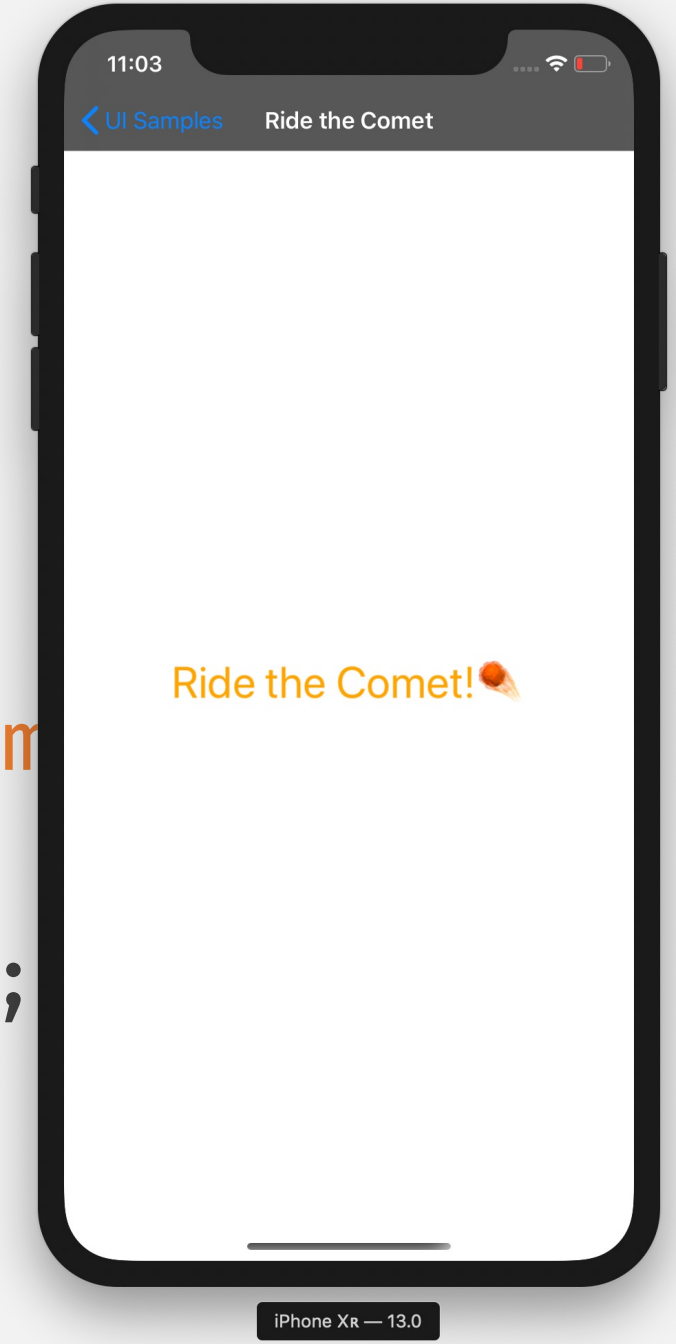
Comet


ButtonSample1.cs — comet
ButtonSample1.cs
sample > Comet.Samples > ButtonSample1.cs
1 using System;
2 using System.Collections.Generic;
3
4 namespace Comet.Samples
5 {
6 public class ButtonSample1 : View
7 {
8 readonly State<int> count = 0;
9
10 [Body]
11 View body() => new VStack
12 {
13 new Button("Increment Value", () => count.Value ++),
14 new Text(() => $"Value: {count.Value}"),
15 };
16
17 }
18 }
19


CreditCardView.cs
CreditCardView ▸ No selection
using System;
namespace HotUI.Samples
{
public class CreditCardView : View
{
}
}
Source
Log
Application Output - HotUI.iOS.Sample
Immediate
Errors
Tasks
Breakpoints
Watch
Call Stack
Name
File
Loaded assembly: /Users/clancey/Library/Developer/CoreSimulator/Devices/2D96258E-20B0-4A1F-8118-96B0ECC3667E/data/Containers/Bundle/Application/05B32964-61A
Loaded assembly: /Users/clancey/Library/Developer/CoreSimulator/Devices/2D96258E-20B0-4A1F-8118-96B0ECC3667E/data/Containers/Bundle/Application/05B32964-61A
2019-07-24 19:51:03.422479-0700 HotUI.iOS.Sample[22259:50198735] SecTaskLoadEntitlements failed error=22 cs_flags=200, pid=22259
2019-07-24 19:51:03.422888-0700 HotUI.iOS.Sample[22259:50198735] SecTaskCopyDebugDescription: HotUI.iOS.Sample[22259]/0#-1 LF=0
Connecting to IDE at tcp://10.104.140.11:9988
Thread started: <Thread Pool> #3
Thread started: <Thread Pool> #4
Thread started: <Thread Pool> #5
Thread started: <Thread Pool> #6
Thread started: <Thread Pool> #7
Thread started: <Thread Pool> #8
Thread started: <Thread Pool> #9
Thread started: <Thread Pool> #10
Thread started: #11
Active View Count: 1
Resolved pending breakpoint at 'ReflectionExtensions.cs:46,1' to bool HotUI.Reflection.ReflectionExtensions.SetDeepPropertyValue (object obj, string name, ol
There is no ViewHandler for HotUI.Samples.CreditCardView
Start receiving updates from ide
Using Mono.CSharp.Evaluator
Loaded assembly: eval-0 [External]
7:51
Toolbox
Unit Tests
Properties
Document Outline

Building the View

```
public class RidePage : View
{
    View body()
        => new Text("Ride the Com
            .FontSize(32)
            .Color(Color.Orange);
}
```



Flow of Data

```
public class RidePage : View
{
    readonly Comet comet;

    View body()
        => new VStack {
            new Text(()=> $"{comet.Rides} rides taken."),
            new Button("Ride the Comet!☄️", ()=>{
                comet.Rides++;
            })
        };
}
```

Single Source of Truth

```
public class Comet : BindingObject
{
    public int Rides {
        get => GetProperty<int>();
        set => SetProperty<int>(value);
    }
}
```

Features

- Model-View-Update
- Fluent Code
- Platform Agnostic
- Property Mapping
- Control Toolkit for Blazor

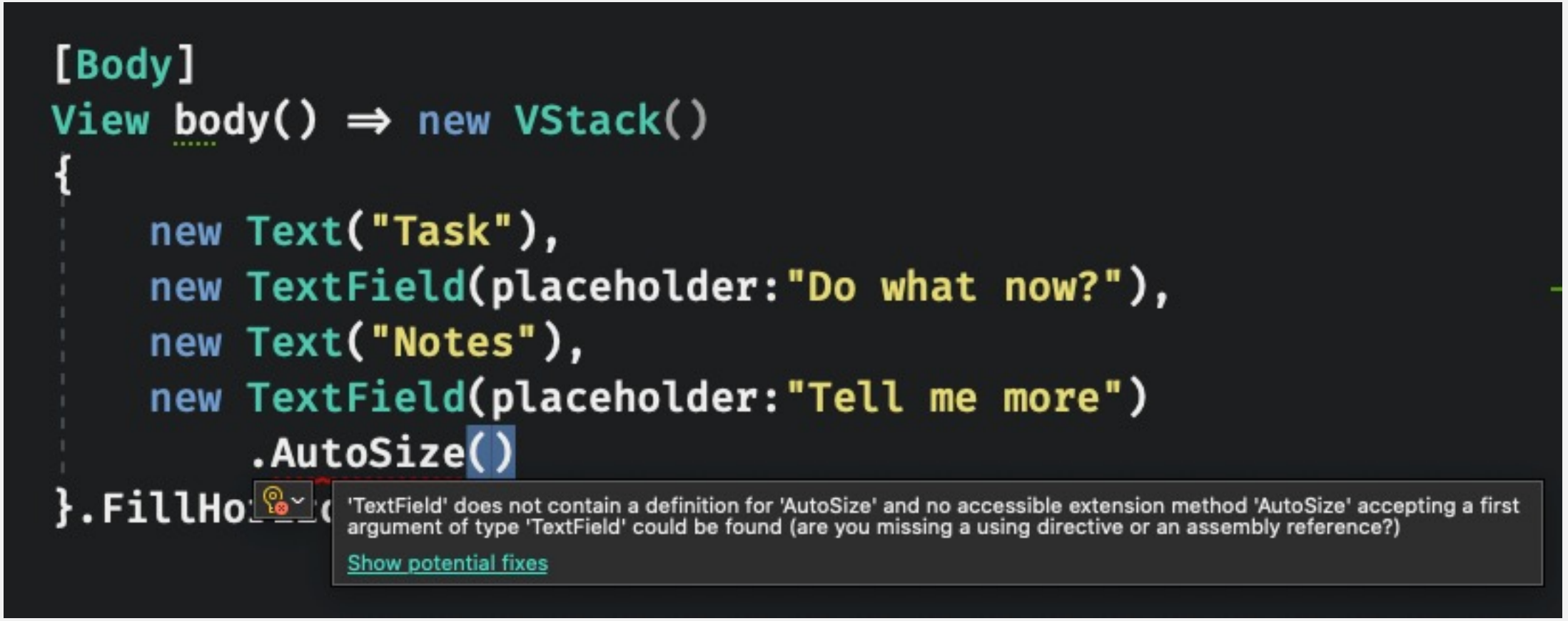
- Can Target:
  - Xamarin.Forms
  - Xamarin.Android
  - Xamarin.iOS
  - Blazor
    - SignalR
    - WebAssembly
  - UWP
  - WPF
  - Xamarin.macOS

Comet Demo

Customizing and Contributing

Comet is architected to be extremely simple to extend, override, and hopefully contribute to.

Example: Not Implemented 😢



Solution: Implement It 🤓

```
public static class LineBreakModeExtensions
{
    public static LineBreakMode GetLineBreakMode<T>(this T view, LineBreakMode defaultMode)
        where T : View
    {
        var mode = view.GetEnvironment<LineBreakMode?>(EnvironmentKeys.LineBreakMode.Mode);
        return mode ?? defaultMode;
    }

    public static T LineBreakMode<T>(this T view, LineBreakMode mode) where T : View
    {
        view.SetEnvironment(EnvironmentKeys.LineBreakMode.Mode, mode);
        return view;
    }
}
```

Solution: Implement It 🤓

```
public class TextHandler : AbstractControlHandler<Text, UILabel>
{
    public static readonly PropertyMapper<Text> Mapper
        = new PropertyMapper<Text>(ViewHandler.Mapper)
    {
        [nameof(Comet.Text.Value)] = MapValueProperty,
        [nameof(EnvironmentKeys.Text.Alignment)] = MapTextAlignmentProperty,
        [EnvironmentKeys.Fonts.Family] = MapFontProperty,
        [EnvironmentKeys.Fonts.Italic] = MapFontProperty,
        [EnvironmentKeys.Fonts.Size] = MapFontProperty,
        [EnvironmentKeys.Fonts.Weight] = MapFontProperty,
        [EnvironmentKeys.Colors.Color] = MapColorProperty,
        [EnvironmentKeys.LineBreakMode.Mode] = MapLineBreakModeProperty,
    };
```

Solution: Implement It 🤓

```
public static void MapLineBreakModeProperty(IViewHandler viewHandler, Text virtualView)
{
    var nativeView = (UILabel)viewHandler.NativeView;
    var mode = virtualView.GetLineBreakMode(DefaultLineBreakMode);
    nativeView.LineBreakMode = mode.ToUILineBreakMode();
    if(mode == LineBreakMode.WordWrap || mode == LineBreakMode.CharacterWrap)
        nativeView.Lines = 0;
    virtualView.InvalidateMeasurement();
}
```

Microsoft

Let's Discuss!

David Ortinau

Principal Program Manager

Twitter: davidortinau

GitHub: davidortinau

Twitch: davidortinau

James Clancey

Principal PM Architect

Twitter: @JTClancey

GitHub: Clancey

Twitch: Clancey